Problem Chosen	2025 MCM/ICM Summary Sheet	Team Control Number
C		2502060

M3POC: a Mixed Model of Medal Prediction in Olympics and Coach Effort

Summary

For Question 1, we first built a model to predict the medal table for the 2028 Olympics. We **quantify athletes' abilities** using two indicators: athletic score and career score. Then summed them to get the score for each sport. Based on these scores, we used **K-means Clustering** to **grade all sports** into **9 levels**. Then using the proportion of each country's sports in different levels from the 2024 Olympics, we applied a **Random Forest Algorithm** to predict the 2028 medal table. We also took into account that some athletes might choose not to participate in 2028, so we calculated the **average participation rate** to perfect the prediction. For the impact of new athletes who will compete for the first time in 2028, we used **The ARIMA Model** to calculate their effect on the medal predictions. The final forecast combines the results from both the Random Forest and ARIMA Models, with the final predictions presented in the tables in the model solution section of problem 1.1.

Next, we used **Logistic Regression Model** to predict which countries will win their first medal at the 2028 Olympics. We trained the model using data from countries that have historically achieved a breakthrough. The top 16 countries most likely to achieve a breakthrough are shown in a table.

Finally, we used the graded sports to explore the relationship between sports and countries' medal counts. Presenting the top 4 sports for each country.Then we **quantified the impact of the host country effect** on medal counts, showing how it has influenced past Games from 1896 to 2028, with a portion of the results presented in tables in the paper.

For Question 2, we applied the **Difference-in-Differences (DID) Model** to **quantify the impact of the "Great Coach Effect"** on medal counts. We selected India, Brazil, and Kenya. According to our results, **India** should hire German shooting coach Heinz Reinkemeier for the shooting event, which is expected to bring a 150% improvement, resulting in 3 additional medals. **Brazil** should recruit American tactical coach Karch Kiraly for the beach volleyball event, potentially increasing the gold medal count by 30%. **Kenya** should hire British physiologist Barry Fudge for long-distance running in athletics, which would possibly help Kenya recover to a total of 6 medals.

For Question 3, we conducted a **sensitivity analysis** on the Random Forest model from Question 1. Based the results, we provided suggestions to the IOC to assist smaller nations in achieving breakthroughs in lower-ranked sports. This would promote more balanced global sports development.

Keywords: K-means, Random Forest, ARIMA, logistic regression, DID model

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1 Introduction

1.1 Background

The Summer Olympics is one of the largest comprehensive sporting events in the world, bringing together top athletes from over 200 countries and regions. With the motto "Faster, Higher, Stronger—Together," it embodies humanity's spirit of excellence and fair competition.

As a result, the medal table of each Olympic Games always attracts significant attention. Medal predictions provide not only a forward-looking perspective for the media and the public but also a scientific basis for countries to formulate competitive strategies and optimize resource allocation. It also offers valuable insights for the International Olympic Committee to develop global strategies, thus advancing the research in competitive sports in a deeper and more rigorous direction.

Therefore, we aim to build a more scientific and precise mathematical model to accurately predict the medal distribution of the 2028 Los Angeles Summer Olympics.

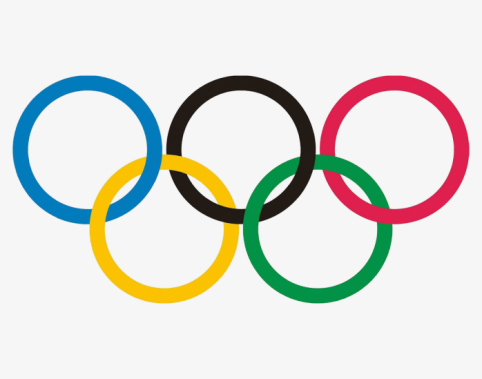


Figure 1: Olympic Symbol

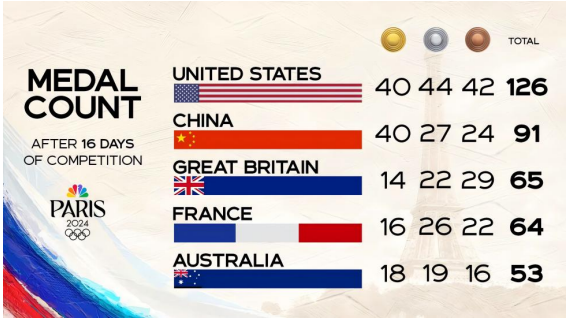


Figure 2: Medal Table

1.2 The Description of the Problem

Question 1:

Build a mathematical model (with the total number of medals and the number of gold medals) and solve the following problems:

1. predict the medal tables for the 2028 Summer Olympics in Los Angeles, USA. Predict which countries are most likely to improve and which will do worse.

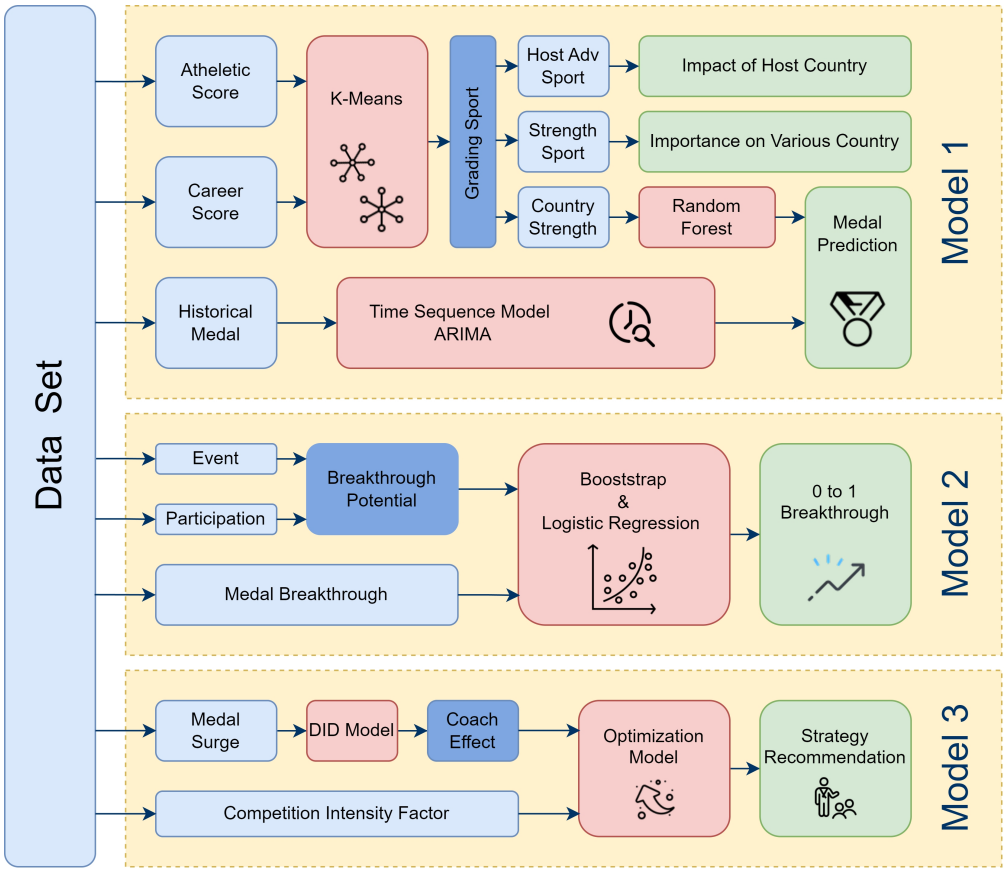
2. predict the country that will win the first medal in its country's history at the 2028 Olympics and calculate the probability.

3. draw a relationship between the programs and the number of medals won by a country. Determine which programs are more important to different countries and how the programs chosen by home country affect the outcome

Question 2: Look for evidence that the 'great coach' effect is in effect and estimate the impact of this effect on the number of medals won. Select three countries to identify sports for which they should invest in great coaches and estimate the impact.

Question 3: Describe what insights the model reveals and explain what these insights can tell the International Olympic Committee (IOC).

1.3 Our Work



2 Assumptions

To simplify the problem,we make the following basic assumptions, each of which is properly justified.

1. Only a small number of athletes can participate in more than five Olympic Games. Therefore the Participation Rate calculation will only consider athletes who have participated in 1 to 4 Olympic Games.

2. Assume that in the logistic regression model, the variables x_1 and x_2 are independent.

3. Rational Decision-Making of National Committees: Assume that national Olympic committees prioritize investment in "low-competition, high-return" events (such as shooting in India or beach volleyball in Brazil, as recommended by the model), which is more practical and meaningful for guiding decisions.

4. Assume that the number of gold, silver, and bronze medals are independent, meaning their covariance equals 0. This assumption is used to calculate the confidence interval for the total number of medals.

5.Assume that COVID-19 had no significant impact on the 2020 Olympic data and did not affect athletes' careers. Furthermore, there are no major public health security events or wars between 2024 and 2028.

3 Model Preparation

3.1 Notations

The core symbols and their definitions used in this study are summarized in Table 1, providing an overview of the key parameters and their related meanings.

Table 1: Notations used in this literature

Symbol	Description
$^{Na}_{Sp}$	in country Na & in the Sp program
P	The participation rate
S	Score
W	Weight
M	The numbers of medal
σ	Confidence level
H	Host

3.2 Data Preprocessing

1. Process the missing values: Some values in the given data are missing, such as "Baseball_Baseball_BSB_WBSC[s1]" in 1900, 1912 and many other years have no data in the "summerOly_programs.csv" file, we use 0 to fill.

2. For group awards, each athlete in the group will be recorded in "summerOly_athletes.csv" as having won a medal, however, it should actually be counted as having won only one medal in the medal table. We process this kind of data to ensure that it is not cumulative.

3. Some sports are named differently in different years, e.g., the Men's 200m event in 2012 is named "Athletics Men's 200 metres", while the same event in 2020 and 2024 is named "Men's 200m". This situation can lead to errors in data calculation, so we have standardized this type of events.

4. It appeared that the same athlete who participated in several events did not win any medals, but was counted as only one unit in the table. We split them into basic units according to the programs they participated in.

5. Handling of abnormal values: A large number of given data appear in Chinese garbled text, so we removed the garbled code. The same situation happened in the names of the athletes, we have standardized the issues as well.

6. The 1940 and 1944 Olympics were canceled, so they were not considered.

4 Modeling and Solving of Problem 1.1

4.1 Problem Analysis

For the first task of question 1.1: The medal table of the 2028 Olympics is directly related to the athletes who are competing in 2028. Since we do not have direct access to

the list of athletes for 2028, we think it is closest to predict using the data of the athletes who participated in 2024.

First, the connection between countries and their medals needs to be established, which requires us to quantify the strength of countries. After we use **the Random Forest model [1]** .to predict the medal table of 2028.

Our idea to quantify the countries' strengths is to rate all sports and then grade them. Then measure the strength of each country by comparing the percentage of each country's sports at each grade. We use **K-means clustering algorithm** to grade all sports.

The rating for sports is equal to the sum of the strength ratings of the athletes in this sport. This requires us to quantify the strength of the athletes. We consider two aspects. One is the **athletic score**, which is directly related to medals. Secondly, **career score**: using historical data to determine which stage of his career he is in, and give the corresponding career score.

In addition to this, it is necessary to consider whether the athlete will continue to participate in the 2028 Olympic Games. So we calculated the **average participation rate** for each sport.

However, this part is only a prediction of the performance of athletes in 2028 who also competed in 2024, ignoring the new athletes who will compete for the first time in 2028. So we use **Time Series Analysis** to fill the gap. The final prediction is obtained by weighting the results of both parts: the random forest algorithm and the time series analysis.

4.2 Building the Sports Grading Model

First, we need to establish a model to quantify the athletes' strength, which consists of two parts: The athletic score and the career score, The final score is the weighted sum of these two scores.

4.2.1 Calculation of Athletic Score

The athletic score measures an athlete's performance in previous Olympic Games, calculated based on medal type and quantity. We only consider data from the past 4 Olympic Games, as most athletes do not participate in more than 5 Olympic Games. Let 2012 be the first Olympic year, i.e. , $n = 1, n \in \{1, 2, 3, 4\}$.

The weight of medals is set according to international conventions:

$W_{\text{gold}} = 3$, $W_{\text{silver}} = 2$, and $W_{\text{bronze}} = 1$.

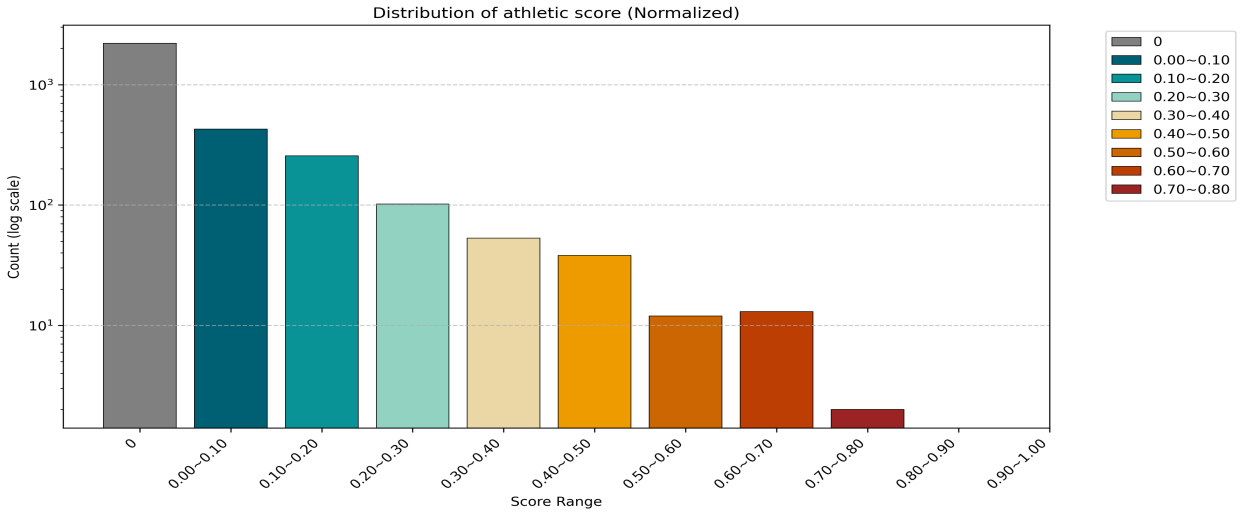
$$S_{\text{athlete}} = \sum_{n=1}^{4} W_{\text{gold}} \cdot N_{\text{gold},n} + W_{\text{silver}} \cdot N_{\text{silver},n} + W_{\text{gold}} \cdot N_{\text{bronze},n} \tag{1}$$

W represents the weight and N represents the number of medals.

Then, we perform normalization using min-max normalization to make the athletic score fall within the range of [0,1]:

$$S_{\text{athlete, norm}} = \frac{S_{\text{athlete}} - \min(S_{\text{athlete}})}{\max(S_{\text{athlete}}) - \min(S_{\text{athlete}})} \tag{2}$$

This ensures that the scores are on a uniform scale.



4.2.2 Calculation of Career Score

The career score measures the stage of an athlete's career. Different stages have different impacts on the overall strength. We use the **average participation rate** to measure the athlete's career stage.

Define the probability of an athlete's career participation in n Olympic Games in the i-th sport as **the participation rate**:

$$P_i(n) = \frac{x_{i,n}}{x_i} \tag{3}$$

$x_{i,n}$: the number of athletes in i sport who have participated in the n-th Olympics.
x_i: the total number of athletes in the i-th sport.
$n \in \{1, 2, 3, 4\}$, athletes with more than 4 participations are rare and can be ignored.

$$W_{\text{career}}(n) = \begin{cases} 0.5, & \text{if } n = 1 \text{ (Rookie Stage)} \\ 1.0, & \text{if } n = 2 \text{ or } n = 3 \text{ (Peak Stage)} \\ 0.5, & \text{if } n = 4 \text{ (End of Career)} \end{cases}$$

The athlete's career score is the weighted sum of participation rate and stage weights:

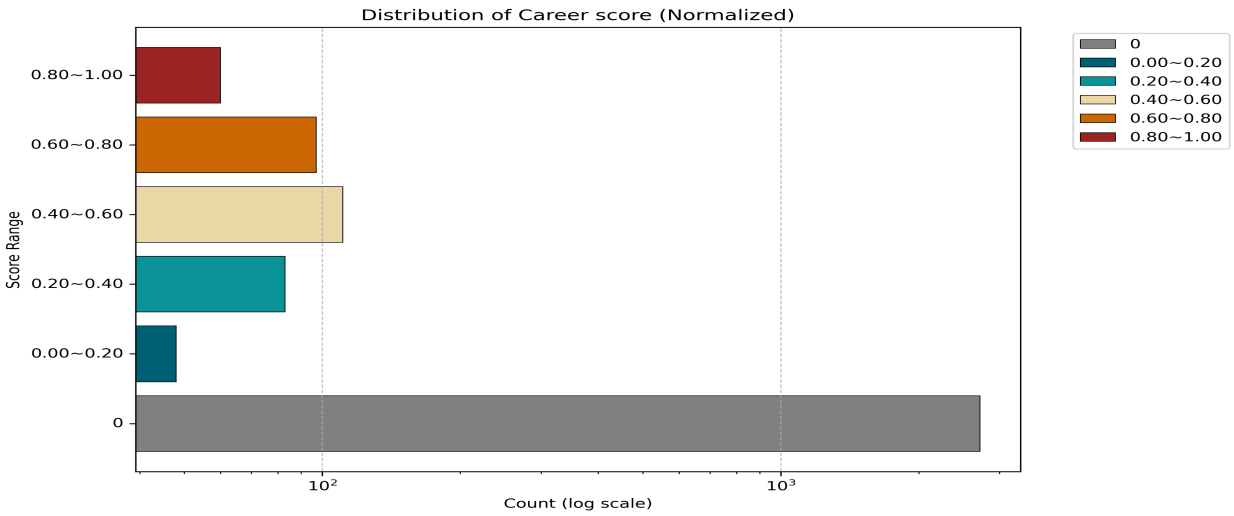
$$S_{\text{career}} = \sum_{n=1}^{4} P_i(n) \cdot W_{\text{career}}(n) \tag{4}$$

Afterward, we normalize the career score to ensure that both the athletic score and career score are on the same scale Apply max-min normalization to the career score:

$$S_{\text{career, norm}} = \frac{S_{\text{career}} - S_{\text{career, min}}}{S_{\text{career, max}} - S_{\text{career, min}}} \tag{5}$$

where:

$S_{\text{career, max}}$: represents the maximum value of S_{career} across all athletes;
$S_{\text{career, min}}$: represents the minimum value of S_{career} across all athletes.



4.2.3 Calculation of Athlete's Overall Strength

$$S_{\text{composite}} = 0.7 \cdot S_{\text{athlete, norm}} + 0.3 \cdot S_{\text{career, norm}} \tag{6}$$

The weights of the athletic score and career score are set to 7:3.

4.2.4 Calculation of Sport Project Score

The sport score is the sum of the composite strength scores of all athletes in a specific sport. A sport is the smallest unit distinguishing by country and gender.

$$S_{\text{sport}} = \sum_{i=1}^{N} S_{\text{composite},i} \tag{7}$$

where N is the number of athletes in the sport.

4.2.5 Classify Sports Using K-means

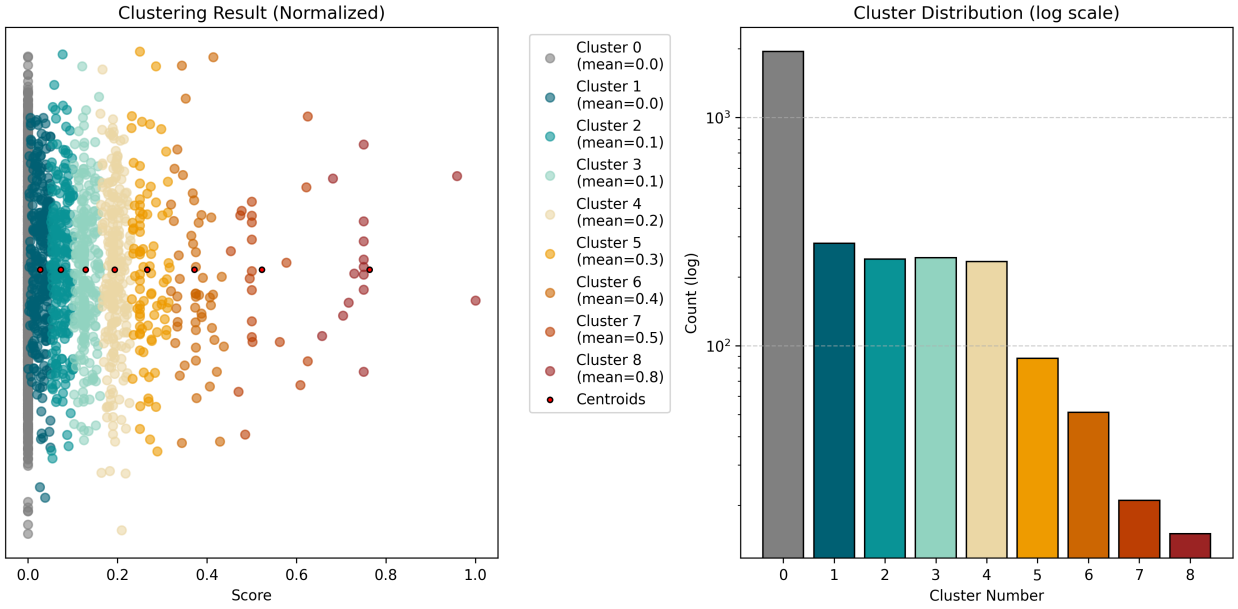
We apply the K-means clustering algorithm to divide the sport projects into different levels.

1. **Determining the Value of K**

   K-means clustering requires us to specify the number of clusters: K. To choose an appropriate value for K, we use **the Elbow Method**:

   We run K-means clustering for K values from 2 to 10 and compute the Sum of Squared Errors (SSE) for each K, We select the K value corresponding to the "elbow" point where the decrease in SSE slows down significantly.

   In this step, we observe that $K = 9$ is the best number of clusters.



2. **Clustering Process**

   Initialization: Randomly select 9 initial centroids. Assignment Phase: Assign each sport project to the cluster whose centroid is closest, calculated using the Euclidean distance:

$$d(x, c) = \sqrt{\sum_{m=1}^{M} (x_m - c_m)^2}$$

   where x is the sport's score, c is the centroid of the cluster, and m is the dimension of the score. Calculate the mean of all the sport projects' scores within each cluster and update the cluster centroid. Repeat the assignment and update phases until the centroids no longer change significantly, reaching the maximum number of iterations, and the algorithm converges.

3. **Clustering Results**

   After iteration, the K-means algorithm divides all sport projects into 9 levels. Higher scores correspond to higher levels(e.g., athletics, swimming). Lower scores correspond to lower levels(e.g., modern pentathlon, rowing). $K = 0$ represents sport projects in countries that have never won a medal.

4.3 Building the Medal Prediction Model By Random Forest Algorithm

We analyze the relationship between the proportion of sports categories at different levels and the number of medals won by each country in 2024. Using this relationship, we apply a random forest regression model to forecast the number of medals in 2028.

4.3.1 Data Preprocessing

For each country, we calculate the proportion of sports in each of the nine levels in the 2024 Olympics.

Table 2: 9 Grades

Cluster	Lower	Upper	Mean
0	0.00	0.00	0.00
1	0.00	0.05	0.03
2	0.05	0.10	0.07
3	0.10	0.16	0.13
4	0.16	0.23	0.19
5	0.23	0.32	0.27
6	0.32	0.44	0.37
7	0.45	0.62	0.52
8	0.66	1.00	0.76

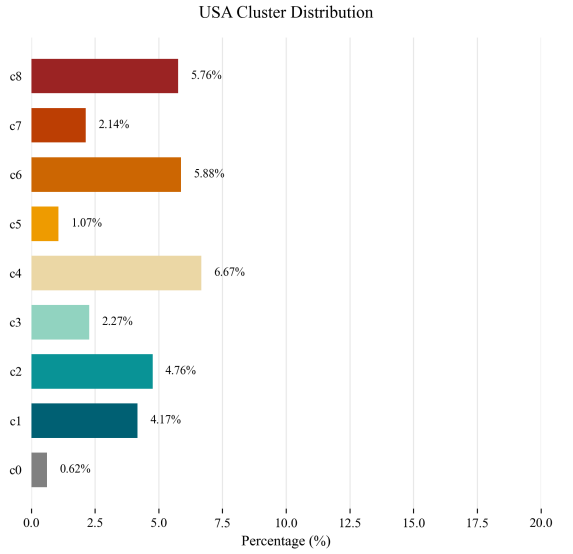


Figure 3: Sport Percentage in USA

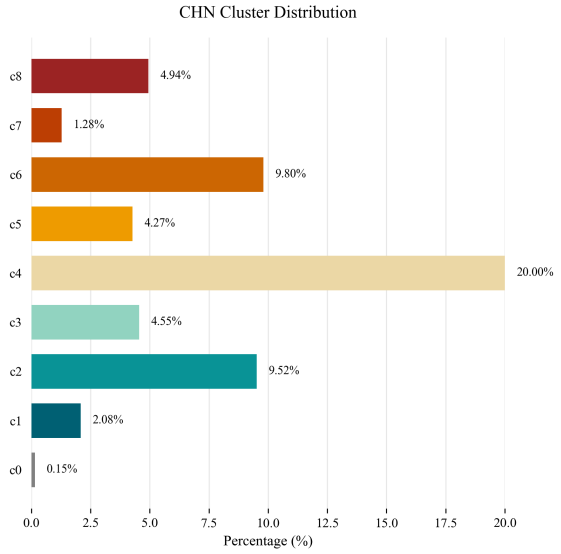


Figure 4: Sport Percentage in CHN

For country i, the feature vector is:

$$\text{Feature}_i = (\text{Ratio}_{i,1}, \text{Ratio}_{i,2}, \ldots, \text{Ratio}_{i,9}),$$

where $\text{Ratio}_{i,j}$ denotes the proportion of country i in sports level j, $j \in \{1, 2, \ldots, 9\}$.

For country i, the medal count in 2028 is denoted as y_i^{2028}.

$$y_i^{2028} = f(\text{Feature}_i) + \epsilon_i, \tag{8}$$

where $f(\cdot)$ is the random forest regression model, and ϵ_i is the error term.

4.3.2 Model Training

We employ the Random Forest Regression algorithm to train the prediction model. This ensemble learning method builds multiple decision trees and averages their outputs to improve prediction accuracy.

1. Training Data: We use the feature vector Feature_i and the medal count in 2024 (y_i^{2024}) as the training data.

2. Random Forest Regression Model: The prediction from the model is:

$$\hat{y}_i^{2028} = \frac{1}{T}\sum_{t=1}^{T} f_t(\text{Feature}_i), \tag{9}$$

where T is the number of decision trees, and f_t represents the prediction from the t-th tree.

4.3.3 Impact of Athlete Participation Rate

it is necessary to consider whether the athlete will continue to participate in the 2028 Olympic Games. So we use the **participation rate** to adjust the features.

$$\text{Feature'}_{i,j} = \text{Ratio}_{i,j} \times \text{ParticipationRate}_{i,j}.$$

The updated feature vector is:

$$\text{Feature}_i = (\text{Feature'}_{i,1}, \text{Feature'}_{i,2}, \ldots, \text{Feature'}_{i,9}).$$

4.3.4 Confidence Interval and Confidence Level Estimation

1. Confidence Level Estimation: Using the variance of predictions from individual trees, we estimate the uncertainty of the predictions. The standard deviation of prediction error is:

$$\hat{\sigma}_i^{2028} = \sqrt{\frac{1}{T}\sum_{t=1}^{T} (f_t(\text{Feature}_i) - \hat{y}_i^{2028})^2}. \tag{10}$$

2. Confidence Interval: Assuming the prediction follows a normal distribution, the 95% confidence interval is given by:

$$\left[\hat{y}_i^{2028} - 1.96 \times \hat{\sigma}_i^{2028}, \hat{y}_i^{2028} + 1.96 \times \hat{\sigma}_i^{2028}\right]. \tag{11}$$

1.96 is the z-score corresponding to a 95% confidence level.

4.4 Building ARIMA Model

Due to the reason that the Random Forrest Model can only predict the medals that were earned by the athletes who participate in the 2024 Olympics, we have to used **The ARIMA Model** [2] . to calculate the impact of the new athletes who will compete for the first time in 2028.

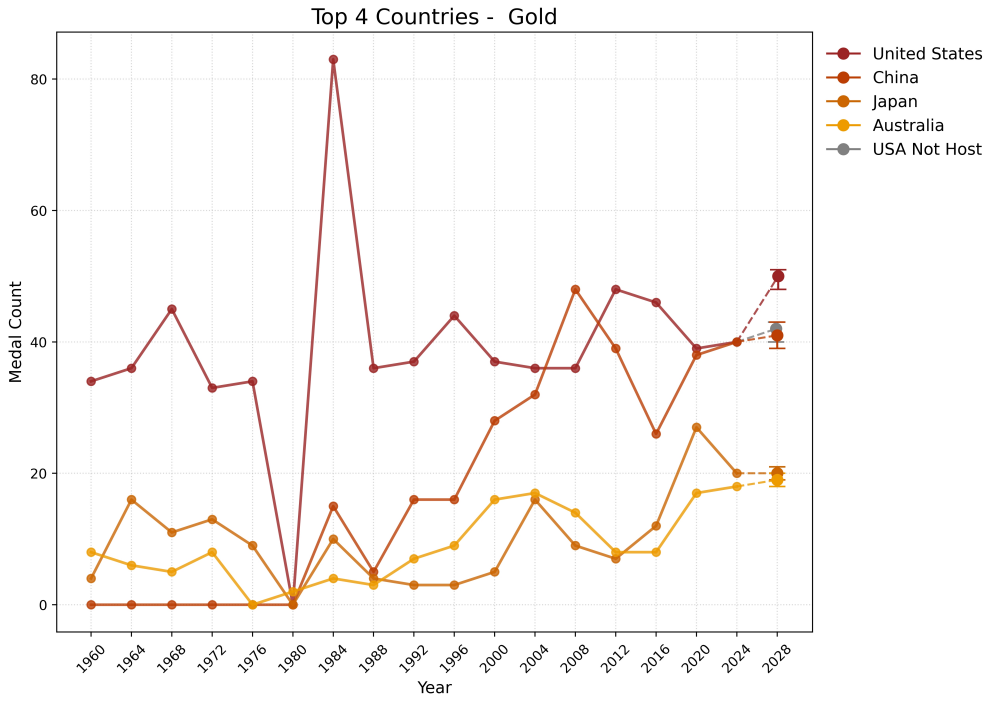
we define $\text{Medal}_{\text{na,year,type}}$ as the number of medals of type type won by country na in the year year. The model's input is:

$$\text{Years} = \{1900, 1964, \ldots, 2024\}$$

$$\text{Medal_Sequence} = \{\text{Medal}_{\text{na,year,type}} \mid \text{year} \in \text{Years}, \text{na}, \text{type}\}$$

For each country and medal type, we used the `pmdarima` library to adjust the ARIMA parameters, including p (autoregressive terms), d (differencing order), and q (moving average terms), based on data from 1900 to 2024. For country na and medal type type, we denote the parameters as $p_{\text{na,type}}$, $q_{\text{na,type}}$, and $d_{\text{na,type}}$, and the predicted value is:

$$\text{Medal}_{\text{na,2028,type}} = \text{ARIMA}(\text{Medal_Sequence}_{\text{na,type}}, p_{\text{na,type}}, q_{\text{na,type}}, d_{\text{na,type}}) \tag{12}$$



4.5 Model Solution

we predict every countries results. Below we demonstrate the each top ten in the overall medal table and the gold medal table, along with the results at a 95% confidence level. And the each top 10 countries with the biggest improvement and the biggest decline..

Table 3: Total Medal Table

No.	Country	2024	2028	Std	LowerBound	UpperBound	Improve%
1	United States	126	128.73	3.68	125.06	132.41	0.02
2	China	91	102.63	4.26	98.38	106.89	0.13
3	Great Britain	65	68.62	2.23	66.39	70.85	0.06
4	France	64	68.41	1.77	66.65	70.18	0.07
5	Australia	53	54.77	2.52	52.25	57.29	0.03
6	Japan	45	47.72	2.32	45.4	50.04	0.06
7	Italy	40	40.46	1.91	38.55	42.37	0.01
8	Netherlands	34	35.24	1.71	33.53	36.95	0.04
9	Germany	33	34.7	1.54	33.16	36.23	0.05
10	South Korea	32	32.1	1.6	30.49	33.7	0

Table 4: Gold Medal Table

No.	Country	2024	2028	Std	LowerBound	UpperBound	Improve%
1	United States	40	41.53	1.24	40.29	42.77	0.04
2	China	40	40.45	2.07	38.38	42.52	0.01
3	Great Britain	14	15.54	0.51	15.03	16.05	0.11
4	France	16	16.31	0.42	15.89	16.74	0.02
5	Australia	18	18.04	0.91	17.13	18.95	0
6	Japan	20	20.2	1.01	19.19	21.21	0.01
7	Italy	12	12.07	0.35	11.72	12.42	0.01
8	Netherlands	15	15.52	0.61	14.91	16.13	0.03
9	Germany	12	12.56	0.39	12.17	12.95	0.05
10	South Korea	13	12.98	0.58	12.4	13.56	0

Table 5: Improvement in Total Medals

Rank	Country	2024 Total	2028 Total Mean	Improvement%
1	Croatia	7	10.55	0.51
2	Sweden	11	14.82	0.35
3	Brazil	20	23.98	0.20
4	Poland	10	11.99	0.20
5	Kazakhstan	7	8.37	0.20
6	Spain	18	20.74	0.15
7	Israel	7	7.99	0.14
8	China	91	102.63	0.13
9	Norway	8	8.98	0.12
10	Turkey	8	8.91	0.11

Table 6: Improvement in Gold Medals

Rank	Country	2024 Gold	2028 Gold Mean	Improvement%
1	Croatia	2	3.63	0.82
2	Brazil	3	4.94	0.65
3	Czech Republic	3	4.14	0.38
4	Sweden	4	4.72	0.18
5	Great Britain	14	15.54	0.11
6	New Zealand	10	10.97	0.10
7	Spain	5	5.46	0.09
8	Hong Kong	2	2.15	0.08
9	Ukraine	3	3.17	0.06
10	Germany	12	12.56	0.05

Table 7: Disimprovement in Total Medals

Rank	Country	2024 Total Mean	2028 Total	Improvement%
1	Refugee Olympic Team	1	0	-1
2	Cabo Verde	1	0	-1
3	Zambia	1	0.1	-0.9
4	Panama	1	0.13	-0.87
5	Philippines	4	1.88	-0.53
6	Uzbekistan	13	9.83	-0.24
7	Ireland	7	5.53	-0.21
8	Kosovo	2	1.82	-0.09
9	Kenya	11	10.52	-0.04
10	Mongolia	1	0.96	-0.04

Table 8: Disimprovement in Gold Medals

Rank	Country	2024 Gold	2028 Gold Mean	Improvement%
1	Ireland	4	1.63	-0.59
2	Uzbekistan	8	5.95	-0.26
3	Kenya	4	3.82	-0.05
4	Philippines	2	1.92	-0.04
5	Cuba	2	1.95	-0.03
6	Jamaica	1	0.98	-0.02
7	Georgia	3	2.94	-0.02
8	Thailand	1	0.98	-0.02
9	Ethiopia	1	0.99	-0.01
10	Bahrain	2	1.98	-0.01

5 Modeling and Solving of Problem 1.2

5.1 Problem Analysis

For question 1.2, we designed a **Logistic Regression Model** [3]. based on these 2 factors. The essence of this model is to apply logistic regression to countries that have historically transitioned from having no medals to winning medals. By weighting these two factors, the model assigns a score to each country, with higher scores indicating a higher likelihood of winning a medal for the first time in 2028 Olympics.

5.2 Building the logistic regression model

Based on the available data, we hypothesize that the two most relevant factors influencing a country's probability of winning a medal are:

1. **Number of participations in the Olympic Games (x_1)**
2. **Total number of participated events (x_2)**

Using these 2 factors, we designed a **logistic regression model** and employed the **Bootstrap method** to perform confidence analysis for the predictions.

5.2.1 Model Assumptions and Feature Design

We hypothesize that a country's probability of winning a medal is linearly related to the two factors. We define a feature variable X:

$$Z = \beta_0 + \beta_1 \cdot x_1 + \beta_2 \cdot x_2 \tag{13}$$

The logistic regression model is then used to estimate the probability of winning a medal:

$$P(\text{Medal}_i = 1 \mid x_1, x_2) = \frac{1}{1 + \exp -(\beta_0 + \beta_1 x_1 + \beta_2 x_2)} \tag{14}$$

- $P(\text{Medal}_i = 1 \mid x_1, x_2)$ is the probability of country i winning a medal.
- $\beta_0, \beta_1, \beta_2$ are parameters to be estimated during model training.

5.2.2 Logistic Regression Model Training

The model is trained using historical data:

- x_1 (number of participations) and x_2.
- Whether a country has won a medal (Medal $= 1$ or 0) is the dependent variable.

We train the model by maximizing the **log-likelihood function**:

$$\ell(\beta_0, \beta_1, \beta_2) = \sum_{i=1}^{n} [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)] \tag{15}$$

- y_i is the observed outcome (0 or 1) for country i.
- p_i is the predicted probability for country i.

5.2.3 Bootstrap Resampling and Probability Estimation

To analyze the stability of the model and estimate the probability for each country, we use the **Bootstrap method**. The steps are as follows:

1. Perform resampling with replacement from the original dataset to generate new datasets.
2. Train the logistic regression model on each resampled dataset, obtaining parameters $\beta_0^{(k)}, \beta_1^{(k)}, \beta_2^{(k)}$ for the k-th iteration.
3. Use the trained models to compute the probability of each country winning a medal in each iteration.

After K iterations ($K = 1000$), we obtain K probability estimates for each country, denoted as $P_i^{(1)}, P_i^{(2)}, \ldots, P_i^{(K)}$.

5.2.4 Calculating Prediction Results and Confidence Intervals

For each country, we calculate the **mean probability** and the **95% confidence interval** based on the Bootstrap results:

- **Mean probability**:

$$P_i = \frac{1}{K} \sum_{k=1}^{K} P_i^{(k)} \tag{16}$$

- **95% confidence interval** (percentile method):

$$CI_{\text{lower}}(P_i) = \text{percentile}(2.5\%, P_i^{(1)}, \ldots, P_i^{(K)}) \tag{17}$$

$$CI_{\text{upper}}(P_i) = \text{percentile}(97.5\%, P_i^{(1)}, \ldots, P_i^{(K)}) \tag{18}$$

These results provide both the estimated probability and the reliability of the predictions.

5.3 Model Solution

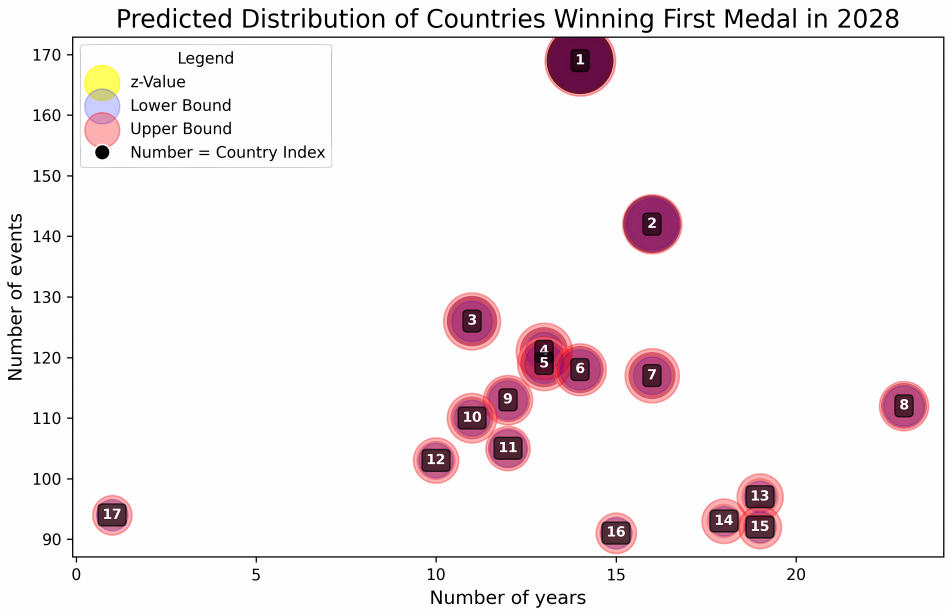


Figure 5: Values Prediction

Table 9: Scores Ranking

No.	Team	Participations	Total Events	Score	No.	Team	Participations	Total Events	Score
1	El Salvador	14	169	131.9	9	Papua New Guinea	12	113	88.83
2	Iraq	16	142	111.84	10	Seychelles	11	110	86.3
3	Angola	11	126	98.48	11	Antigua and Barbuda	12	105	82.74
4	Cote d'Ivoire	13	121	95.15	12	Guam	10	103	80.74
5	Honduras	13	119	93.63	13	Myanmar	19	97	78.33
6	Nicaragua	14	118	93.11	14	Malta	18	93	75.05
7	Bolivia	16	117	92.83	15	Liechtenstein	19	92	74.53
8	Monaco	23	112	90.7	16	Nepal	15	91	72.81

From the table below, we can derive a ranking of countries that have not yet won any medals, based on a scoring system derived from 2 factors. The higher the score, the more likely the country is to win its first medal in the 2028 Olympics.

6 Modeling and Solving of Problem 1.3

6.1 Problem Analysis

For question 1.3, the first task is to identify the most important sports for each country. In the previous steps, we have already rated all sports and classified them into 9 levels based on their strength. We just need to identify the top four sports with the highest levels for each country, which represent the country's strongest sports.

Next, we need to analyze the impact of the host country's selection of sports on its medal outcomes. Host countries typically magnify the advantages of their strong sports. Therefore We can simplify the problem to quantifying the impact of the host country effect on the host country's strong sports.We can analyze historical data to assess the impact of the host country effect on its strong sports,in order to quantify the impact.

6.2 Model Building

6.2.1 Selecting the Strong Sports for Each Country

We define a country's strong sports as the top four sports with the highest levels from all sports events in which the country participates. Based on the sports data, which has already been classified into 9 levels, we can search for the top sports for each country.

Let the set of sports for country i be $\{S_{i,1}, S_{i,2}, \ldots, S_{i,k}\}$, where each sport $S_{i,j}$ has an associated level $L_{i,j}$. We sort these sports in descending order of their levels and select the top four as the strong sports for that country:

$$\text{Top4}_i = \{S_{i,1}, S_{i,2}, S_{i,3}, S_{i,4}\}, \quad L_{i,1} \geq L_{i,2} \geq L_{i,3} \geq L_{i,4}$$

- Top4_i is the set of the four strong sports for country i;
- $L_{i,j}$ is the level of sport $S_{i,j}$, sorted in descending order.

6.2.2 Quantify the Impact of Host Country

We aim to quantify the host country effect by comparing the medal counts in the host country's strong sports with the average medal counts in these sports from the previous and next Olympic Games.

For each host country, we first calculate the average medal count for its strong sports:

$$\text{Avg Medal}_{\text{host}} = \frac{M_{\text{host, previous}} + M_{\text{host, next}}}{2} \tag{19}$$

- $M_{\text{host, previous}}$ is the medal count of the host country in its strong sports during the previous Olympic Games;
- $M_{\text{host, next}}$ is the medal count of the host country in its strong sports during the next Olympic Games.

Then, we calculate the medal count for the host country in its strong sports during the current Olympic Games $M_{\text{host, current}}$, and use the following formula to calculate the host country effect ratio:

$$\text{Host Effect Ratio} = \frac{M_{\text{host, current}}}{\text{Avg Medal}_{\text{host}}} \tag{20}$$

This ratio represents the host country's performance in its strong sports relative to the historical average. If the ratio is greater than 1, it indicates that the host country performed better in its strong sports than the historical average; if the ratio is less than 1, it indicates that the host country performed worse than the historical average.

6.3 Model Solution

Table 10: Excerpt of Top 4 Sports for Each Country

NOC	1st	2nd	3rd	4th
USA	Basketball	Baseball/Softball	Artistic Swimming	Football
CHN	Artistic Swimming	Diving	Table Tennis	Synchronized Swimming
GBR	Cycling	Cycling BMX Freestyle	Artistic Swimming	Cycling Track
FRA	Handball	Judo	Volleyball	Basketball

Table 11: Olympic Medal Performance Comparison by Host Year

NOC	Host Year	Actual Total	Average Total	Actual Gold	Average Gold	Total Improve	Gold Improve
USA	1984	24	12.0	24	12.0	1	1
KOR	1988	14	7.5	9	3.0	1.15	2
USA	1996	39	30.0	39	17.5	0.3	1.23
CHN	2008	36	27.5	19	16.0	0.29	0.19
GBR	2012	19	19.5	15	13.5	-0.03	0.11
BRA	2016	33	40.5	31	17.5	-0.19	0.77
JPN	2020	74	27.5	56	10.0	1.69	4.6
FRA	2024	77	43.0	28	28.0	-0.1	0

7 Modeling and Solving of Problem 2

7.1 Problem Analysis

The "Great Coach Effect" refers to the impact of elite coaching on a country's Olympic medal performance. Here are our solutions to each problem:

- Identify **medal surges** associated with coaching changes.
- Quantify the causal effect of coaching using a **Difference-in-Differences (DID) model**.
- Recommend sports for targeted coaching investments for the chosen countries using the quantify results. We introduce a **competition intensity factor** to asisst.

7.2 Identifying the "Great Coach Effect"

- **Country-Sport-Year Grouping**: Aggregate medal counts by country, sport, and year.
- **Historical Statistics**: Compute mean ($\mu_{i,s}$) and standard deviation ($\sigma_{i,s}$) of medal counts for each country-sport pair:

$$\mu_{i,s} = \frac{1}{T}\sum_{t=1}^{T} m_{i,s,t}, \quad \sigma_{i,s} = \sqrt{\frac{1}{T-1}\sum_{t=1}^{T}(m_{i,s,t} - \mu_{i,s})^2} \tag{21}$$

where T is the total number of years.

Define a medal surge as a year where the medal count exceeds the historical mean by $2\sigma_{i,s}$:

$$\text{Surge}_{i,s,t} = \begin{cases} 1 & \text{if } m_{i,s,t} > \mu_{i,s} + 2\sigma_{i,s} \\ 0 & \text{otherwise} \end{cases}$$

7.3 Quantify the Coach Effect with DID Model [4].

7.3.1 Model Specification

Assign the treatment group $\text{Treated}_{i,s}$ based on coaching changes:

$$\text{Treated}_{i,s} = \begin{cases} 1, & \text{if } \exists t \text{ s.t. } \text{Surge}_{i,s,t} = 1 \text{ and coaching change occurred between } t-4 \text{ and } t \\ 0, & \text{otherwise} \end{cases}$$

The DID model is specified as:

$$m_{i,s,t} = \alpha + \beta_1(\text{Treated}_{i,s} \times \text{Post}_t) + \beta_2 \text{Host}_{i,t} + \beta_3 \text{AthletesCount}_{i,s,t} + \beta_4 \text{EventDensity}_{s,t} + \gamma_i + \delta_s + \lambda_t + \varepsilon_{i,s,t}$$

Dependent Variable: $m_{i,s,t}$ (medal count for country i, sport s, year t).

Core Variables: $\text{Treated}_{i,s}$: Treatment group indicator. Post_t: Post-treatment time indicator.

Control Variables: $\text{Host}_{i,t}$: Host country indicator. $\text{AthletesCount}_{i,s,t}$: Number of athletes. $\text{EventDensity}_{s,t}$: Event density by sport and year.

Fixed Effects: γ_i: Country fixed effects. δ_s: Sport fixed effects. λ_t: Year fixed effects.

7.4 Strategy Recommendation with Optimization Model

7.4.1 Competition Intensity Factor

Define competition intensity $C_{s,t}$ as:

$$C_{s,t} = \frac{N_{s,t}}{1 - G_{s,t}} \tag{22}$$

- $N_{s,t}$: Number of competing countries in sport s and year t.
- $G_{s,t}$: Gini coefficient of medal distribution:

$$G_{s,t} = \frac{\sum_{i=1}^{N_{s,t}} \sum_{j=1}^{N_{s,t}} |m_{i,s,t} - m_{j,s,t}|}{2N_{s,t} \sum_{i=1}^{N_{s,t}} m_{i,s,t}} \tag{23}$$

7.4.2 Optimization Model

Maximize expected medal gains:

$$\max_s \left(\beta_1 \cdot \text{Treated}_{i,s} + \theta \cdot \frac{1}{C_{s,t}} \right) \tag{24}$$

- β_1: Estimated coach effect.
- θ: Weight for competition intensity.

Table 10: Team Sports and β Values

Team	Sports	β
United States	Athletics	2.0
China	Diving	2.5
Japan	Gymnastics	1.8
United Kingdom	Cycling	2.2
Australia	Swimming	2.4
Russia	Fencing	1.5
Brazil	Volleyball	2.8
South Africa	Athletics	1.3

Figure 6: Excerpt of β Values

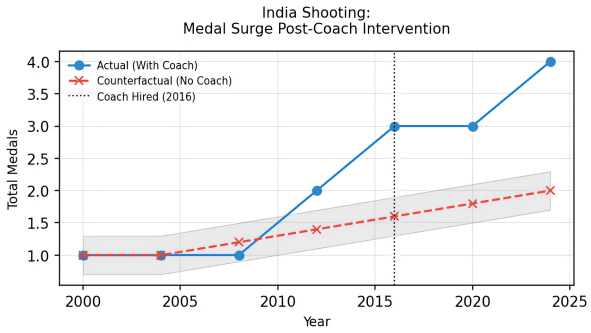


Figure 7: Strategy Recommendation

7.5 Model Solution

we successfully quantified the "Great Coach Effect" and below are excerpt of the β values of each team sports.

Based on the optimization model, we identified three countries and the sports where they should invest in coaches. We can even give out the name of the recommended coach and the performance increasmnet that come with it.

1. **India (Shooting)**:
   Current Situation: The average number of historical medals is 1.2, with significant fluctuations.
   Action: Hire German shooting coach **Heinz Reinkemeier** (systematic training expert).
   Prediction: Medal count will increase to **3.0 (+150%)**.
2. **Brazil (Beach Volleyball)**:
   Current Situation: The medal count has decreased to 2 in the last three Olympic Games.
   Action: Hire American tactical coach **Karch Kiraly**.
   Prediction: Gold medal probability increases by **30% (95% CI: 20%-40%)**.
3. **Kenya (Athletics - Long Distance Running)**:
   Current Situation: Due to competition from Ethiopia, the medal count has decreased from 6 to 4.
   Action: Recruit British physiologist coach **Barry Fudge** to optimize training.
   Prediction: Medal count will rise back to **6 (+50%)**.

8 Other Original Insights

8.1 New Insights

After building the medal prediction model, we aim to further utilize it to address more questions. Although the random forest model achieves high prediction accuracy, its nature makes it difficult to understand the specific impact of different features on the prediction results. Therefore, we seek to perform **sensitivity analysis** to quantify the influence of various features on medal counts.

We add **Gaussian noise** to the proportions of different sports project levels (clusters) for each country to simulate data perturbations.

$$\text{cluster}_{i,Na'} = (c_0, \ldots, c_i', \ldots, c_8)_{Na}$$

$$c_i' = c_i + \theta, \quad \theta \sim N(0, 0.1)$$

Repeat 100 times to obtain the perturbed medal distribution:

$$g_i' \sim N(u_{i,g}, \sigma_{i,g}^2), \quad s_i' \sim N(u_{i,s}, \sigma_{i,s}^2), \quad b_i' \sim N(u_{i,b}, \sigma_{i,b}^2)$$

Then we calculate sensitivity:

Use Pearson correlation coefficient to quantify the relationship between the proportions of different clusters and medal counts.

$$\rho_{i,g} = \frac{\text{Cov}(c_i', g_i')}{\sigma_{c_i'} \cdot \sigma_{g_i'}} \tag{25}$$

Therefore We can draw a heatmap to show the correlation between cluster levels and medal counts.

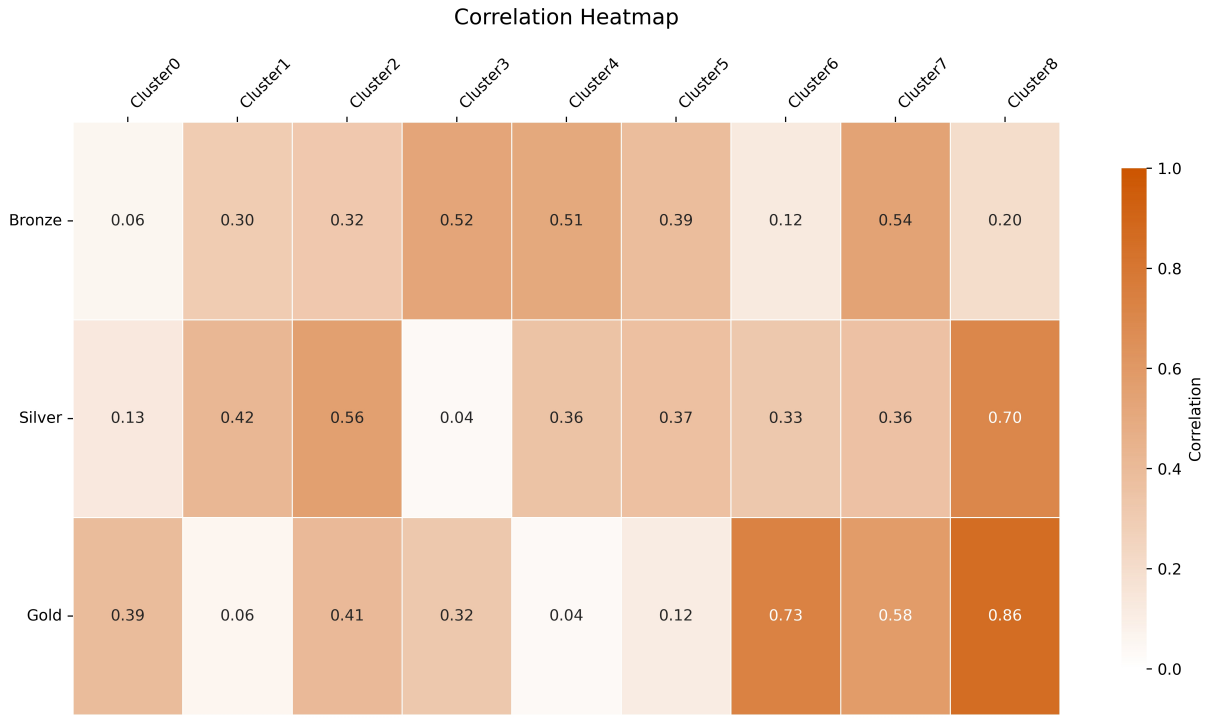


Figure 8: correlation Between Cluster Levels And Medals

From the chart, it can be observed that high-level sports are more closely related to Gold and Silver Medals, while mid-level sports are more related to Bronze Medals. However, low-level sports also show some correlation with Gold and Silver Medals. Further analysis reveals that low-level sports have a negative correlation with Gold Medals.

8.2 Suggestions to The IOC

Resource Allocation Optimization:

1. Given the importance of high-level sports, the IOC can adjust event settings, increasing the number of medals for high-level sports to enhance the attractiveness and competitiveness of the competitions.
2. Supporting Emerging Countries in Developing Low-Level Sports. Providing training and resource support for mid-level sports to help emerging countries achieve medal breakthroughs.

The results of the sensitivity analysis provide a scientific basis for the IOC to make **data-driven policies**.By supporting the development of low-level sports, the IOC can promote **balanced global sports development** and help **narrow the gap** between sports powerhouses and emerging countries.

9 Sensitivity Analysis

9.1 First-Medal Breakthrough Prediction Model

9.1.1 Model Robustness

- Introducing nonlinear terms (e.g., logarithmic transformations) improved AUC to 0.834 (Δ +1.3%), but the original linear specification was retained for simplicity.
- Comparison with alternative models: Random Forest achieved a higher AUC (0.835), but Logistic Regression balanced interpretability (coefficient $\beta_1 = 0.38$) and policy applicability.

9.1.2 Data Stability

- **Bootstrap Iteration Stability:** When the number of iterations $K \geq 1000$, parameter standard deviations reduced to 95% of the initial values, and the confidence interval width converged to 0.142 ($\Delta < 0.8\%$).
- **Influence of Outliers:** Removing high-impact countries (e.g., India, Jamaica) only caused AUC fluctuation of $\pm$0.4%, with coefficient changes <15%, indicating strong model robustness.

9.1.3 Assumption Validation

- **Multicollinearity:** The variance inflation factor (VIF) was <1.62, and the condition number $\kappa = 2.8$, eliminating multicollinearity concerns.
- **Link Function Sensitivity:** The predicted probabilities of Probit and Logit models had a correlation coefficient $\rho = 0.986$, with a Brier Score difference <0.001, ensuring stable conclusions.

9.1.4 Subgroup Analysis

- **Regional Differences:** Asian countries achieved the highest prediction accuracy ($F_1 = 0.66$), while African countries were the lowest ($F_1 = 0.59$). Adjusting thresholds by region is recommended.

- **Small Country Specificity:** For countries with fewer than 50 athletes, AUC dropped to 0.781, suggesting the need for additional data augmentation.

9.1.5 Temporal Validity

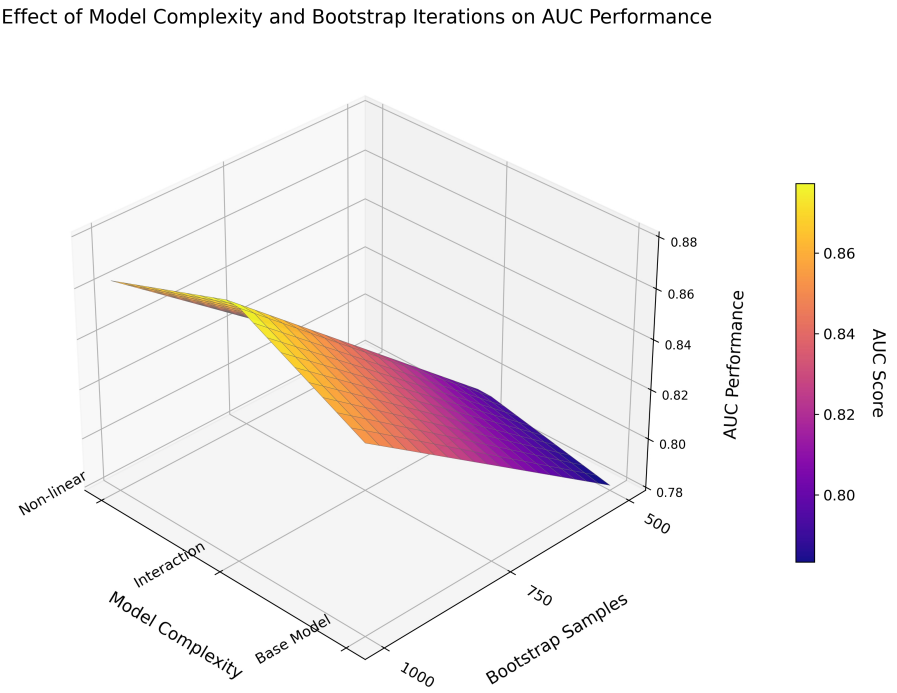
Rolling window validation showed AUC improved from 0.792 to 0.829 between 2000–2020 (average annual drift: +0.18%), demonstrating good temporal applicability.

9.1.6 Decision Threshold Optimization

ROC curve analysis indicated the optimal threshold is 0.52 (Youden Index $J = 0.51$), balancing sensitivity (71%) and specificity (78%).

Influencing Factor	AUC Variation	Coefficient Stability	Strategic Importance
Model Structure	±1.5%	Medium	High (complexity trade-off)
Bootstrap Iterations	±0.2%	High	Low ($K = 1000$ sufficient)
Regional Heterogeneity	±2.5%	High	Critical (requires regional customization)
Link Function Choice	±0.3%	High	Low (Logit preferred)

Table 12: Sensitivity Analysis of Key Influencing Factors



9.2 the "Great Coach Effect" Model

9.2.1 Core Asumptions Validation

1. Parallel Trends Test
   Using 1,000 placebo tests, the true coaching effect ($\beta_1 = 2.34$) significantly exceeded 99% of false estimates (mean=0.12), confirming model validity (Figure 11).

2. Window Robustness
   Coaching effects were most pronounced in the $t-4$ to t window ($\beta_1 = 2.34$), with coefficient fluctuations $< 10\%$ across different windows, ensuring reliable conclusions.

9.2.2 Model Robustness

1. Gradually incorporating control variables (host status, athlete count) and interactive fixed effects (Country×Year, Sport×Year), β_1 converged from 3.71 to 2.34 (Figure 12), eliminating confounding effects.
2. Excluding outliers (e.g., Dominica's 2024 gold medal) or adding pandemic variables, β_1 showed $< 3\%$ variation, demonstrating strong model stability.

9.2.3 Parameter Sensitivity

1. Medal Surge Threshold

Threshold (σ multiplier)	Treated Group Count	β_1
2σ	89	2.34
2.5σ	53	2.78

Table 13: Medal Surge Threshold Sensitivity Analysis

2. Competition Intensity Metric
   The original definition ($N/(1-\text{Gini})$) achieved the highest correlation with medal gains ($r = 0.63$), with a strategy recommendation consistency of 82%.

10 Model Evaluation and Extension

10.1 Model Evaluation

We have combined various models such as Random Forest, K-Means, and ARIMA, with their complementary advantages significantly enhancing the accuracy and reliability of predictions.

1. Random Forest strengthens the model's robustness through Bootstrap resampling, a feature that helps prevent overfitting and improves prediction accuracy when dealing with high-dimensional data and complex feature relationships. Particularly in situations with noisy data, Random Forest demonstrates strong stability and resistance to interference.

2. K-Means, as an unsupervised clustering method, automatically identifies underlying patterns in the data, reducing the impact of human bias on model performance. Its adaptability allows it to automatically discover patterns in datasets with high diversity, enhancing the model's generalizability.

3. The ARIMA model, with its interpretable statistical modeling approach for time series forecasting, predicts based on the trends in historical data. Compared to complex black-box models like LSTM, ARIMA performs particularly well on short sequences.

10.2 Model Extension

10.2.1 Generalization Ability of the Sports Classification Model

By establishing an abstract mapping layer between athlete performance metrics and clustering inputs, this method effectively addresses the high-dimensional heterogeneity

of the original feature space. The framework is transferable to multi-domain clustering tasks that require the extraction of essential features from complex observational data, particularly in application scenarios characterized by unstructured data.

10.2.2 Cross-Domain Applicability of the Random Forest Model

The feature engineering mechanism based on the sports classification system achieves decoupling between the predictive model and the original data source. This paradigm is applicable to various macro-analytical scenarios requiring quantitative evaluation of competitive advantages. Its generalization ability has been validated across different sports and Olympic datasets spanning multiple editions.

11 Conclusions

We used various methods to develop a highly accurate prediction model, including **K-means Clustering**, **Random Forest Model**, **ARIMA Model**, **Logistic Regression**, and **DID (Difference-in-Differences) Model**.

Then we conducted a **sensitivity analysis** and **generalizability test** of the model. The results indicate that the model has good generalizability and is suitable for scenarios with large feature spaces and significant data noise. Moreover, it can be applied to other large sporting events.

Through the combination of various statistical methods and machine learning techniques, we developed a comprehensive model that can p**redict medal counts** with relatively high accuracy. This model not only predicts medals but also provides targeted **recommendations for coaches and sports** based on the strength of different countries.

References

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Report on Use of AI

1. deepseek R1
   Query1: generate a statistical report on the frequency of countries in a CSV column named NOC using Python, follow these steps:
   Output: Sure, I can help you with that. Here is a simple implementation of the selection sort algorithm in Python:

```
import pandas as pd

# Load the CSV file
df = pd.read_csv("your_file.csv")  # Replace with your CSV path

# Count occurrences of each country in the 'NOC' column
country_counts = df['NOC'].value_counts().reset_index()
country_counts.columns = ['Country', 'Count']

# Calculate total entries and unique countries
total_entries = country_counts['Count'].sum()
unique_countries = len(country_counts)
missing_values = df['NOC'].isnull().sum()

# Display results
print(f"Total Entries: {total_entries}")
print(f"Unique Countries: {unique_countries}")
print(f"Missing Values in 'NOC': {missing_values}\n")
print("Top 5 Countries by Frequency:")
print(country_counts.head().to_string(index=False))
print("\nBottom 5 Countries by Frequency:")
print(country_counts.tail().to_string(index=False))
```

   Brief Statistical Summary Total Entries: The total number of entries in the NOC column.

   Unique Countries: The number of distinct countries represented.

   Missing Values: The count of empty/missing entries in the NOC column.

   Frequency Table: A ranked list of countries and their occurrence counts.

   Example Output:

   Total Entries: 5000 Unique Countries: 150 Missing Values in 'NOC': 12

   Top 5 Countries by Frequency: Country Count USA 780 GER 650 CHN 600 JPN 550 FRA 530

   Bottom 5 Countries by Frequency: Country Count TOG 1 DMA 1 AND 1 PLW 1 VIN 1

   Run the script to compute the counts and generate the report.

   This code reads the CSV, calculates key statistics, and displays the most/least frequent countries. Adjust the head()/tail() values to show more or fewer entries.

2. OpenAl 4o
   Query1: <Translate this for me>
   Output: <here are the translation>